Movie Booking System in Java

A console-based movie ticket booking application

Uses Scanner, switch-case, methods, conditions.

Calculates discounts, GST, and the final ticket price

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VA PROJECT

1. Objective of the Program

- Display a list of movies with their ticket prices
- Allow the user to select a movie
- Allow the user to choose a showtime and seat
- Apply applicable discounts and calculate the final bill
- Display booking and payment details

2. Movie Selection

- The program displays four movies and their prices:
- 1. Spiderman Brand New Day – Rs.250
- 2. Avengers Doomsday – Rs.350
- 3. Avengers Secret Wars – Rs.460
- 4. Salaar 2 – Rs.170
- A switch statement identifies the selected movie and price.



CINEMA SYSTEM

3. Showtime Selection

- Two showtimes are available:
- • Morning
- • Evening
- Morning has no time-based discount.
- Evening receives a 10% discount on the movie price.
- Another switch statement handles the selection.

4. Seat Selection

- The user enters a seat number from 1 to 40.
- An if condition checks whether the seat number is valid.
- If valid → booking is confirmed.
- If invalid → the program displays an error and stops.



CINEMA SYSTEM

5. Billing & Discount Logic

- Base price is taken from the selected movie.
- Time discount is added when the evening show is selected.
- The intended design also supports a coupon discount.
- Total discount = time discount + coupon discount.
- Price after discount cannot be less than zero.

6. GST & Final Price

- GST is calculated after discounts.
- GST rate used in the code: 1.4%
- GST = price after discount × 0.014
- Final price = price after discount + GST
- The bill displays the movie, showtime, seat, discounts, GST, and total paid.

7. Java Concepts Used

- Scanner → reads user input
- Variables → store movie, price, seat, and discount information
- switch-case → handles movie and showtime choices
- if-else → validates the seat and prevents negative totals
- String → stores movie names, showtime, and coupon code
- double → handles prices and decimal calculations
- Method call → intended for coupon calculation



JAVA CONCEPTS

8. Program Flow

- Start
- ↓
- Display movies → Select movie
- ↓
- Select showtime → Apply evening discount if applicable
- ↓
- Select seat → Validate seat number
- ↓
- Enter coupon → Calculate discounts
- ↓
- Calculate GST → Calculate final price
- ↓
- Display movie bill → End



SYSTEM FLOW

9. Important Corrections Needed

- The current code contains references to objects/variables that are not declared:
- • applyCoupon(couponCode, movie.basePrice)
- • showtime.discount
- • movie.basePrice
- These will cause compilation errors.
- Also, after nextInt(), nextLine() is needed before reading the coupon input to consume the pending newline.



DEBUG & FIX

10. Conclusion

- The program demonstrates the basic structure of a Java ticket-booking system.
- It combines input handling, decision-making, validation, and arithmetic calculations.
- With the undefined movie/showtime references corrected and the coupon method implemented, it can be made fully executable.
- The system can later be extended with multiple seats, payment options, booking IDs, and stored bookings.